

Anti-jamming Trajectory Design Method of UAV Based on Intelligent Optimization Algorithm

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Abstract—This paper proposed an intelligent optimization algorithm to design the anti-jamming trajectory of the unmanned aerial vehicle (UAV) aided in wireless sensor networks (WSNs). In practical terms, a UAV is used to collect data from a plenty of ground sensor (GSs) while existing multiple of ground jammers (GJs) under line-of-sight (LoS) channel model. We aim to maximum the data collection average rate by optimizing the horizontal trajectory. We provide two different intelligent optimization algorithms to design trajectory of the UAV, Random Iterative Algorithm and Tree seed Optimizer Algorithm (TSA). Simulation result shows that the latter one is better in improving the data collection rate and enhance the security of the network with multiple jammers.

Keywords—UAV; trajectory; Intelligent optimization; Anti-jamming

I. INTRODUCTION

Wireless sensor networks (WSNs) are widely employed in various application scenarios, for instance, smart city, battlefield monitoring, and post-disaster rescue [1]. Unmanned aerial vehicle (UAV) has the advantages of wide coverage, high mobility performance and low power consumption, which possesses great potential for introducing into WSNs to collect the data from sensors [2]. Therefore, UAV-assisted WSNs have attracted extensive attention recently. In [3], a 3D Trajectory optimization method is proposed for UAV-enabled data collecting. Considering the directional antennas instead of an omnidirectional antenna, UAV's altitude and beamwidth are optimized in [4] to acquire the minimal data collection time. From the view of energy effectiveness, a UAV routing is designed with low time-consuming in [5]. Dynamic and heterogeneous WSNs are considered in [6] and the corresponding UAV trajectory optimization method is put forward.

Nevertheless, the UAV-sensors links are line-of-sight (LoS) dominant, which results in serious security threat such as eavesdropping [7] and jamming [8]. When the security of UAV-assisted WSNs is considered, the UAV trajectory optimization problem becomes more complicated. Among these security problems, the Intentional or unintentional interference signals from the ground jammers (GJs) are extremely intractable. By introducing the reinforcement learning, an UAV anti-

jamming method is presented in [9], where the UAV is considered to be static. With the advantage of the mobility of UAV being exploited, an anti-jamming UAV trajectory is given in [10]. To avoid adopting the simplified LoS model, a probabilistic LoS channel is considered in [11], where this model is more practical for some scenarios. However, the contributions mentioned above all take into account a single jammer, which is too idealistic for complicated electromagnetic scenarios, since more than one jammer may exist in these scenarios.

In this paper, we consider the UAV-assisted WSNs under the case with multiple Jammers. The UAV is dispatched to collect data when passing through the GSs successively. At first, the UAV will be assigned initial path at every slot. As the number of iterations continues to increase, the UAV will choose a better position to obtain greater signal-to-noise ratio. In random iterative algorithm, the UAV's position is change randomly. However, in TSA, the UAV's trajectory is based on the growth and reproduction mode of natural trees. Simulation result testify the validity and reliability of TSA.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider the wireless transmissions assisted by a UAV to collect aerial data from S ($S > 1$) GSs, denoted by $G_S = \{GS_1, GS_2, \dots, GS_S\}$. Meanwhile M ($M > 1$) malicious GJs, denoted by $J = \{J_1, J_2, J_3, \dots, J_M\}$, intend to jam the legitimate communications. GSs and GJs are located on the ground with two-dimensional horizontal coordinates $w_s = [x_s, y_s]^T$, $w_j = [x_j, y_j]^T$ respectively.

Assume that the UAV flies at the fixed height H_0 , with the flight period T . In order to facilitate analysis, the flight period T is divided into I equal-time slots, each with sufficient step by $\delta = T/I$. Thus, letting $q[i] = [x_i, y_i]^T$ represent the coordinate of UAV in the i -th time slot where $i = 1, \dots, I$, the trajectory of the UAV can be described by $q[1], \dots, q[I]$ approximately. To ensure the UAV's flight trajectory periodically and reachability in one time slot, two constraints should be satisfied as follows

$$q[1] = q[I] \quad (1)$$

$$\|q[n] - q[n-1]\| \leq D_{\max} \quad (2)$$

Where $D_{\max}$ is the maximum distance that the UAV can travel within a time slot. The power of GS can be defined as $P = [P_1, P_2, \dots, P_S]$, and the GJ with a constant power

$$Q = [Q_1, Q_2, \dots, Q_M] \quad (3)$$

$h_k(i)$ is defined as the legitimate channel gain between the ground sensor k and the UAV at the i -th time slot

$$h_k(i) = \sqrt{\beta_0 d_{ki}^{-\delta}} \quad (4)$$

$g_m(i)$ indicates the channel gain between the jammer m and the UAV

$$g_m(i) = \sqrt{\beta_0 d_{mi}^{-\delta}} \quad (5)$$

where β_0 is the channel gain at a reference distance $d_0 = 1m$, d_{ki} and d_{mi} is the distance from the ground sensor k and the jammer m to the UAV in time slot i respectively. δ represents the average path loss exponents. So we can get the received data in each slot expressed as

$$y(i) = \sqrt{h_k(i)} \cdot \sqrt{P_k} \cdot x + \sum_{m=1}^M \sqrt{g_m(i)} \cdot \sqrt{Q_m} \cdot J_m + n \quad (6)$$

Where x is the transmitted signal by the selected GS_k .

The first part of the formula (5) is the legal signal, and the part II are the interference signals.

Assuming that there is only one legitimate ground sensor sending information to the UAV in each time slot. The rate from the k sensor to the UAV in time slot i can be indicated as

$$R_k(i) = \log_2 \left(1 + \frac{h_k(i) \cdot P_k}{\delta^2 + \sum_{m=1}^M g_m(i) \cdot Q_m} \right) \quad (7)$$

Where δ^2 is the power of the noise received. Then we can select the maximum rate from all the sensor as the designated sensor in time slot i . The data collection rate with UAV trajectory can be approximately formulated as

$$R(i) = \frac{1}{I} \sum_{i=1}^I \max \{R_1(i), R_2(i), \dots, R_S(i)\} \quad (8)$$

III. ANTI-JAMMING UAV TRAJECTORY DESIGN

Because of the UAV trajectory is random and non-convex so it is difficult to find the optimal solution. In this section we will propose two methods to design different trajectory.

A. Random Iterative Algorithm

The first method is a random iterative algorithm which is introduced by the next form.

Algorithm 1 random iterative algorithm to get the trajectory

1: **Initialize** the position of the UAV q which is Uniformly distributed in the circular

$$q_1(:, end) = q_1(:, 1)$$

2: **Repeat**

3: increase random increment to the initial position

$$q_2 = q_1 + (2 * rand(size(q_1)) - 1)$$

The increment is limited to the maximum flying distance of an UAV and the new positions need to satisfy

$$q_2(:, end) = q_2(:, 1)$$

4. Comparing the data rate in new position with the initial data rate, if the data rate increases, then update the position of the UAV.

5. **Until** the iterative times finish.

The random iterative method is the exhaustive method that we need traverse the nearby area in each time slot.

B. The Tree Seed Optimizer Algorithm

The second method is the Tree seed Optimizer Algorithm. It is a new swarm intelligence optimization algorithm based on the growth and reproduction mode of natural trees. First of all, it will produce a batch of trees randomly in the search area by the Eq.(12)

$$T_{i,j} = L_{j,\min} + r_{i,j} (H_{j,\max} - L_{j,\min}) \quad (9)$$

In the equation, the $T_{i,j}$ is the position of the tree, $H_{j,\max}$ and $L_{j,\min}$ represent the upper and lower boundary of the search area respectively. $r_{i,j}$ is a random number between 0 and 1. The trees have different ability to produce seeds, so we should find the optimal position according to the objective function as the Eq.(13)

$$B = \min(f(\vec{T}_i)), i = 1, 2, \dots, N \quad (10)$$

Then the tree will produce the new seed by the tree in the optimal position by two mechanisms as the Eq.(14)

$$S_{i,j} = \begin{cases} iw * T_{i,j} + \alpha_{i,j} * (T_{i,j} - T_{r,j}), & rand > ST \quad (a) \\ iw * T_{i,j} + \alpha_{i,j} * (B_j - T_{r,j}), & otherwise \quad (b) \end{cases} \quad (11)$$

ST is constant. If the random number is larger than ST , we should choose the first equation to guarantee the global search. The global search can avoid the iteration process falling into part of the best. The second equation is the local search which can be conducive to the convergence of the algorithm. S represents the seed, T denotes the tree. They all have multi-dimensional elements B_j is the optimal tree's j -th element in the current location. $\alpha_{i,j}$ is a random number on behalf of the step size factor. The way considers a combination of global and local searches.

The flow chart of the TSA is described as the below form.

Algorithm 2 Tree seed Optimizer Algorithm to get the trajectory

Step 1: The initialization of the algorithm.
Set the number of the population size(N).
Set the ST parameter for the method.
Set the dimensionality of the problem (D).
Decide the termination condition
Generate N random tree location on the D-dimensional search space using the Eq.(12):
Evaluate the tree location using objective function specified for the problem.
Select the best solution using the Eq.(13).

Step 2: Searching with Seeds
For all trees
Decide the number of seeds produced for this tree.
For all seeds
For all dimensions
IF(rand<ST)
Update this dimension using the Eq.(14)(b)
ELSE
Update this dimension using the Eq.(14)(a)
END IF
END FOR
END FOR
Select the best seed and compare it with the tree.
If the seed location is better than tree location, the seed substitutes for this tree.
END FOR

Step 3: Selection of Best Solution
Select the best solution of population.
If new best solution is better than the previous best solution, new best solution substitutes for the previous best solution.

Step 4: Testing Termination condition
If termination condition is not met, go to Step 2.

Step 5: Reporting
Report the best solution.

IV. NUMERICAL RESULTS

In this section we show the simulation result to verify the improved algorithm in elevating the data collection rate by changing the trajectory of the UAV. We consider a UAV-enabled data collection system in a square area. We arrange four GS_s and three GJ_s in this area. The required simulation parameters are set as follows: $\partial = 2.2$, $\beta = 10^{-6}$, $\sigma^2 = 10^{-13}$, $V_{\max} = 50$, $P_s = 1$, $P_j = 1$, $pop = 50$, $dim = 100$, $ST = 0.1$.

Fig.1 compares the trajectory of the TSA in different iterative number, we can see that the route will converge as the iterative number larger. Because the TSA has strong ability to find excellence. Fig.2 shows the trajectory of the UAV by two different algorithms with GS_s and GJ_s

distributed randomly when $T = 50s$. We can obtain from the figure that the UAV will be farer away from GJ_s than GS_s . The route of the TSA is closer than the Random. Then the UAV will cost less energy to collect all data in limited time.

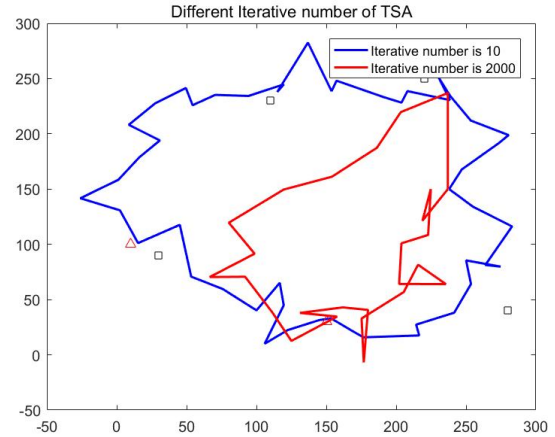


Fig. 1 Different Iterative number of the TSA

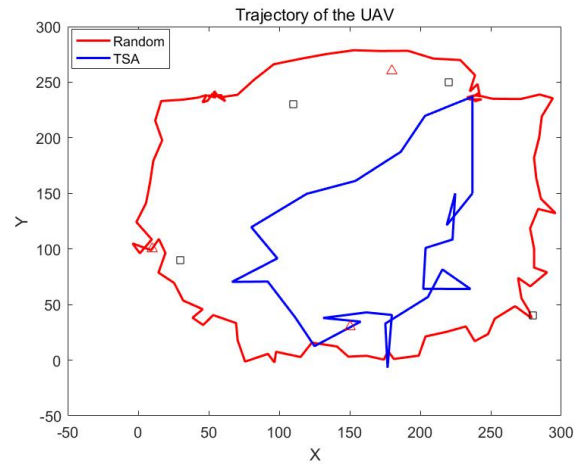


Fig. 2 The Trajectory of the UAV

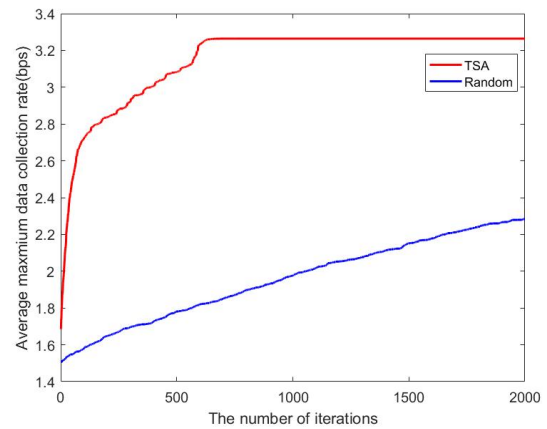


Fig. 3 Average maximum data collection rate

In Fig.3 we analyze the average maximum data collection rate of TSA and random. The TSA has larger

number than random algorithm obviously and the convergence is faster. Hence, the TAS's performance is better than the random algorithm and the anti-jamming ability is better.

V. CONCLUSION

In this paper, we studied the UAV's anti-jamming trajectory in the WSNs network. We optimized the maximize data collection rate by scheduling GS transmission and the UAV trajectory on LOS model. In order to analyze the problem, we propose two methods to iterate. The result shows that TSA algorithm has significant rate performance and convergence speed, which validates the practical significance of the UAV-enabled anti-jamming network.

REFERENCES

- [1] Wei Z, Zhu M, Zhang N, et al. UAV Assisted Data Collection for Internet of Things: A Survey[J]. IEEE Internet Of Things Journal, 9(17):15460-15483, 2022.
- [2] Z. Wei, Z. Feng, H. Zhou, L. Wang, and H. Wu, "Capacity and delay of unmanned aerial vehicle networks with mobility," IEEE Internet Things J., vol. 6, no. 2, pp. 1640-1653, Apr. 2019.
- [3] You C, Zhang R. 3D Trajectory Optimization in Rician Fading for UAV-Enabled Data Harvesting[J]. IEEE Transactions on Wireless Communications, 18(6): 3192-3207, 2019.
- [4] Hyun-Ho Choi, Jung-Ryun Lee. Joint Optimization of Altitude and Beamwidth for UAV-Powered Wireless Sensor Networks[J]. IEEE Transactions on Vehicular Technology, 72(1): 1279-1284, 2023.
- [5] Baek, Jaeuk, S. I. Han, and Y. Han. "Energy-Efficient UAV Routing for Wireless Sensor Networks." IEEE Transactions on Vehicular Technology, 69(2): 1741-1750, 2020.
- [6] Jie Chen and Jianhua Tang. UAV-Assisted Data Collection for Dynamic and Heterogeneous Wireless Sensor Networks[J]. IEEE Wireless Communications Letters, 11(6): 1288-1292, 2022.
- [7] M. Li, X. Tao, N. Li, H. Wu and J. Xu, "Secrecy Energy Efficiency Maximization in UAV-Enabled Wireless Sensor Networks Without Eavesdropper's CSI," in IEEE Internet of Things Journal, vol. 9, no. 5, pp. 3346-3358, 1 March 1, 2022, doi: 10.1109/JIOT.2021.3098049.
- [8] M. Mah, H. Lim, and A. W. Tan, "UAV relay flight path planning in the presence of jamming signal," IEEE Access, vol. 7, pp. 40 913-40 924, Mar. 2019.
- [9] L. Xiao, C. Xie, M. Min, and W. Zhuang, "User-centric view of unmanned aerial vehicle transmission against smart attacks," IEEE Trans. Veh. Technol., vol. 67, no. 4, pp. 3420-3430, Apr. 2018.
- [10] Y. Wu, W. Yang, X. Guan and Q. Wu, "UAV-Enabled Relay Communication Under Malicious Jamming: Joint Trajectory and Transmit Power Optimization," in IEEE Transactions on Vehicular Technology, vol. 70, no. 8, pp. 8275-8279, Aug. 2021, doi: 10.1109/TVT.2021.3089158.
- [11] B. Duo, Q. Wu, X. Yuan and R. Zhang, "Anti-Jamming 3D Trajectory Design for UAV-Enabled Wireless Sensor Networks Under Probabilistic LoS Channel," in IEEE Transactions on Vehicular Technology, vol. 69, no. 12, pp. 16288-16293, Dec. 2020, doi: 10.1109/TVT.2020.3040334.